Ch 5.1.4-5: More Cross-Validation

Lecture 14 - CMSE 381

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Fri, Oct 4, 2024

Announcements

Last time:

- k-fold CV

This lecture:

- More k-fold CV
- Bias-Variance Tradeoff
- CV for classification

Announcements:

- Exam 1 grades
- HW #4 Posted.
  - Changed Deadline! Due Wednesday Oct 9.

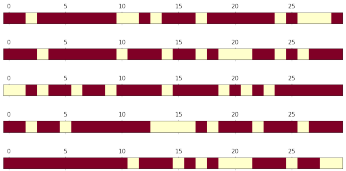
Lec #	Date			Reading	HW
12	Mon	9/30	Leave one out CV	5.1.1, 5.1.2	
13	Wed	10/2	k-fold CV	5.1.3	
14	Fri	10/4	More k-fold CV,	5.1.4-5	
15	Mon	10/7	k-fold CV for classification	5.1.5	
16	Wed	10/9	Resampling methods: Bootstrap	5.2	HW #4 Due Weds 10/9
17	Fri	10/11	Subset selection	6.1	
18	Mon	10/14	Shrinkage: Ridge	6.2.1	
19	Wed	10/16	Shrinkage: Lasso	6.2.2	
20	Fri	10/18	Dimension Reduction	6.3	HW #5 Due Fri 10/18
	Mon	10/21	No class - Fall break		
	Wed	10/23	**Review**		
	Fri	10/25	**Midterm #2**		

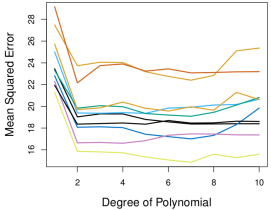
Section 1

k-fold CV

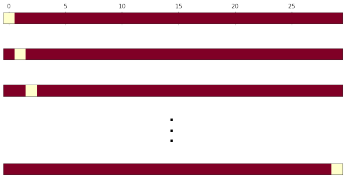
Approximations of Test Error

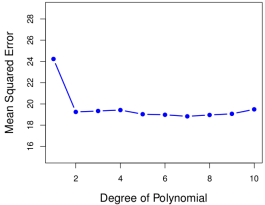
Validation Set



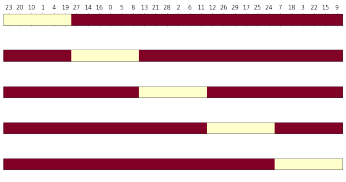


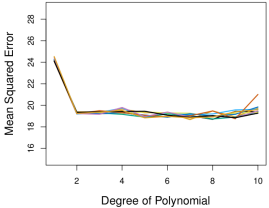
LOOCV





K-fold CV



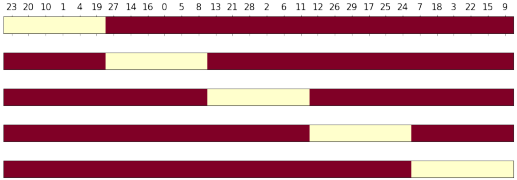


Definition of k-fold CV

- Randomly split data into k-groups (folds)
- Approximately equal sized. For the sake of notation, say each set has ℓ points

- Remove ith fold U_i and reserve for testing.
- Train the model on remaining points
- Calculate $\mathrm{MSE}_i = \frac{1}{\ell}\sum_{(x_j,y_j)\in U_i}(y_j - \hat{y}_j)^2$

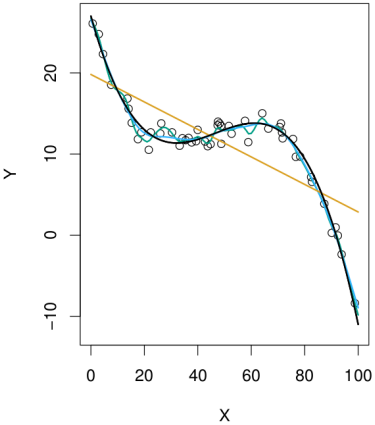
- Rinse and repeat

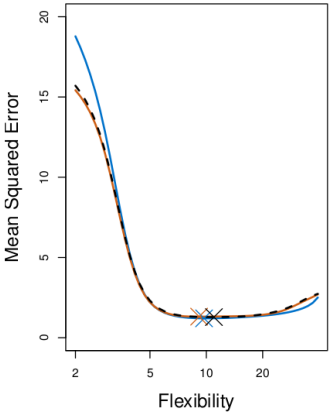


Return

$$CV_{(k)} = \frac{1}{k}\sum_{i=1}^{k}\mathrm{MSE}_i$$

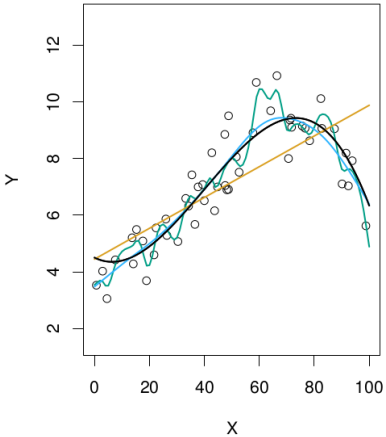
Comparison with simulated data: Ex 3

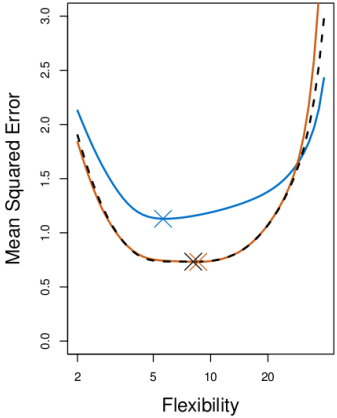




- Blue: true test MSE is shown in blue
- Black dashed: the LOOCV estimate
- Orange: 10-fold CV
- The crosses indicate the minimum of each of the MSE curves.
- Nearly the same

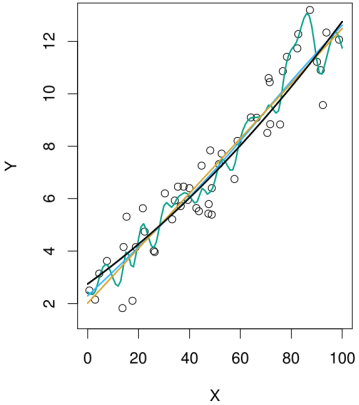
Comparison with simulated data: Ex 1

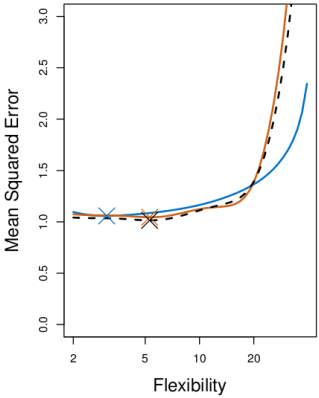




- Blue: true test MSE is shown in blue
- Black dashed: the LOOCV estimate
- Orange: 10-fold CV
- The crosses indicate the minimum of each of the MSE curves.
- **Underestmates the true test MSE**

Comparison with simulated data: Ex 2





Close for lower flexibility, then overestimates at higher ranges

Takeaways from the examples

- In all plots, LOOCV and 10-fold CV are similar
- Sometimes really only care about the minimum point in the test MSE because want to get the right degree of freedom for a model. All examples have approximately same x-coord for that

Bias-Variance Tradeoff: Bias

$$E(y_0 - \hat{f}(x_0))^2 = \mathrm{Var}(\hat{f}(x_0)) + \left[\mathrm{Bias}(\hat{f}(x_0))\right]^2 + \mathrm{Var}(\varepsilon)$$

- 1.... Validation set overestimates test error b/c used small subset of data
- 3.... k-fold gives medium level of bias b/c training set has approximately $(k-1)n/k$ observations
- 2....LOOCV gives approximately unbiased estimate since uses almost all data every time
- soooooooooo LOOCV better than k-fold just for bias

Bias-Variance Tradeoff: Variance

$$E(y_0 - \hat{f}(x_0))^2 = \mathrm{Var}(\hat{f}(x_0)) + \left[\mathrm{Bias}(\hat{f}(x_0))\right]^2 + \mathrm{Var}(\varepsilon)$$

- LOOCV higher variance:
  - avg the outputs of n fitted models, but all on almost identical observations
  - high correlation with each other
- k-fold
  - k fitte models somewhat less correlated with each other
  - mean of many highly correlated quantities has higher variance than those not corrleated

- Empirically, $k = 5$ or 10 has been shown to be a happy medium

Section 2

Using K-Fold CV on Polynomial Linear Regression

Polynomial regression

Replace linear model

$$y_i = \beta_0 + \beta_1 x_1 + \varepsilon_i$$

with

$$y_i = \beta_0 + \beta_1 x_1 + \beta_2 x_1^2 + \cdots + \beta_d x_1^d + \varepsilon_i$$

- Can learn with linear models by passing in predictors x_i^ℓ
- Tend to not go higher than degree 3 or 4 because makes overly flexible

Faking linear regression into doing our work for us

- Draw a column of data
- Draw the version with squares and cubes in the columns
- Pass to linear regression and interpret the coefficients as the ones we want.

Coding - Build a plot for train/test scores vs flexibility

Next time

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